

Instructor: Peter Mao

Office: Korman 209

Email: pm929@drexel.edu

MRC/Office Hours: Thursdays 3:00-5:00

Main Text: *Discrete Mathematics: An Open Introduction*, 4th edition by Oscar Levin

Supplementary Texts: *Number Theory: In Context and Interactive* by Karl-Dieter Crisman

Applied Combinatorics by Mitchel T. Keller and William T. Trotter

Note that all three of these textbooks are meant to be accessed online for full functionality. The main textbook has solutions and/or hints to most of the exercises therein, so they can serve as a source of comparison or inspiration for homework problems. The other two textbooks both take advantage of live code evaluation which can help to make some things more concrete.

Course Description:

Elementary set theory, combinatorics, elementary number theory, graphs, and special topics chosen from formal language theory, graph algorithms, coding theory, and other applications.

Pre-Requisites:

MATH 220 Minimum Grade: C or CS 270 Minimum Grade: D or ECE 200 Minimum Grade: D

How This Course Is Graded (Mastery-Based Testing)

This course uses *mastery-based testing*: your grade is determined by which course learning objectives you can demonstrate mastery of, not by an accumulation of partial credit. The course has 11 **Core** learning objectives and 6 **Advanced** learning objectives. Each is assessed via a short *proficiency check* (PC) on which you must produce essentially correct work to pass. There is no partial credit: a PC is either passed at the mastery level or "not yet" passed.

If you do not pass a PC on your first attempt, you may retake it on a new version. Retakes are taken in the MRC by appointment. Tests are not returned, but you may review your work and a solution key with the instructor before retaking. Mastering an objective at any point during the term (including on the final exam) counts the same as mastering it on the first attempt.

Core and Advanced Objectives

Writing proofs is central to discrete mathematics and is a Core skill of this course — it appears in nearly every Core objective, not only the ones whose names mention proofs. Over the term you will be expected to write element-chasing proofs of set identities (C_1), prove that specific functions are injective or surjective (C_4), justify combinatorial counts rigorously (C_5 – C_7), verify closed forms for recurrences by induction (C_3 , C_8), and derive facts using Bézout's identity (C_9), among others. The Advanced objectives extend this by asking for proofs that combine multiple techniques or that carry a longer chain of reasoning.

The 11 Core objectives (C_1 through C_{11}) are the content that any student passing the course should be able to demonstrate fluency in:

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| 1. Sets, Logic, and Quantifiers (incl. element-chasing proofs) | 7. Counting III — Inclusion–Exclusion |
| 2. Basic Proof Methods (direct, contrapositive, contradiction) | 8. Recurrence Relations (incl. verifying closed forms by induction) |
| 3. Induction (weak and strong) | 9. GCD and the Euclidean Algorithm (incl. Bézout’s identity) |
| 4. Functions and Bijections (incl. proofs of injectivity/surjectivity) | 10. Modular Arithmetic (incl. Fermat’s little theorem) |
| 5. Counting I — Principles and Permutations | 11. Graph Fundamentals (incl. handshake lemma, Eulerian circuits, basic coloring) |
| 6. Counting II — Combinations and Bijective Counting | |

The 6 Advanced objectives (A_1 through A_6) are enrichment that distinguishes higher grades:

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| 1. Combinatorial Proofs and Pigeonhole | 4. Number Theory Proofs |
| 2. Binomial Theorem and Identities | 5. Graph Theory Proofs |
| 3. Generating Functions | 6. Cross-Topic Synthesis |

What kind of proof at each tier? A Core proof is typically short and stays inside one technique — prove a single divisibility statement by induction, or show a function is injective directly from the definition. An Advanced proof typically chains multiple techniques together — e.g., a combinatorial argument that needs both PIE and a bijection, or an induction whose inductive step uses the Euclidean algorithm. The standard for “passing” is the same at either tier (the proof is essentially correct, with minor mistakes at most); what changes is the complexity of argument expected.

A detailed description of what mastery of each objective looks like will be posted to the course site alongside this syllabus, along with practice problems and sample PCs.

Proficiency Checks

PCs for Core objectives are first attempted in class on designated dates announced at least one week in advance. PCs for Advanced objectives open after the relevant material is covered and are taken in MRC at your initiative (schedule an appointment!). A typical PC takes 15–25 minutes.

Number of attempts. Each PC has a base allotment of attempts. Completing the relevant unit’s homework (see below) unlocks additional attempts as follows:

- **Core PCs:** 3 attempts by default; **5 attempts** if you have submitted the relevant unit’s homework.
- **Advanced PCs:** 2 attempts by default; **3 attempts** if you have submitted the relevant unit’s homework.

For each PC retake, a new version of the PC is given. The final exam serves as one additional attempt (not included in the above counts) at any unmastered Core or Advanced PC.

Homework (Formative)

Weekly homework is assigned in two parts:

- **WeBWorK questions** drawn from the textbook, submitted automatically via WeBWorK. You have **unlimited attempts** on each WeBWorK problem.
- **Written questions**, primarily proof-based, submitted via Gradescope. Solutions should be in LaTeX or written digitally; a legible photograph of paper work is also acceptable.

Homework is graded on a complete/incomplete basis: it is your practice for the proficiency checks and is *not* a direct input into the letter grade. A unit's homework counts as **complete** when you have earned **at least 90%** on its WeBWorK set (where one is assigned) and have made a genuine attempt at each of its written problems (where assigned). Completing a unit's homework unlocks additional retake attempts on that unit's PC (see above) — the system's way of recognizing that you have engaged with the material before spending a retake.

You are encouraged to get help on *both* parts of the homework — from your classmates, from me, and from the Math Resource Center in Korman 207. Your written submissions should be in your own words, with your sources and those whom you consulted cited.

Final Exam

The final exam is cumulative and serves two purposes: a last-chance attempt at any Core objective not yet mastered, and an opportunity to demonstrate any Advanced objective. The exam is sectioned by objective, so a student who has already mastered all 11 Core objectives need only attempt sections corresponding to Advanced objectives they wish to demonstrate.

Grade Determination

Your letter grade is determined entirely by how many Core and Advanced objectives you have mastered by the end of the term. The base requirement for a passing grade (*C* or above) is mastering all 11 Core objectives. Each Advanced objective mastered raises the grade by one-third of a letter from that baseline, with A_6 (the cross-topic synthesis capstone) required to reach the *A* range.

Grade	Core mastered	Advanced mastered
A+	11/11	6/6 (necessarily includes A_6)
A	11/11	5/6, including A_6
A–	11/11	4/6, including A_6
B+	11/11	4 or 5/6 without A_6
B	11/11	3/6
B–	11/11	2/6
C+	11/11	1/6
C	11/11	0/6
C–	10/11	—
D+	9/11	—
D	8/11	—
F	$\leq 7/11$	—

The capstone synthesis objective A_6 functions as the gate to the *A* range: 4 or 5 Advanced objectives mastered without A_6 caps at *B+*. Borderline cases are resolved using my professional judgment. The intent of the grade scale is that a *C* means *ready for any course that requires Math 221 as a prerequisite*, and an *A* means *has gone substantially beyond that, including a demonstrated ability to synthesize ideas across topics*.

Schedule of Topics

The course is organized into 10 weeks of instruction plus a final exam week. The detailed week-by-week schedule of PCs will be posted alongside this syllabus. The tentative plan is:

Week	Topics	Objectives introduced
1	Sets, logic, quantifiers	C_1
2	Basic proof methods (direct, contrapositive, contradiction)	C_2
3	Induction; first-order recurrences	C_3, C_8 (part 1)
4	Functions and bijections; counting principles, permutations	C_4, C_5
5	Combinations, stars-and-bars, bijective counting	C_6
6	Inclusion–Exclusion; second-order linear recurrences	C_7, C_8 (part 2)
7	Divisibility, gcd, Euclidean algorithm	C_9
8	Modular arithmetic, linear congruences, Fermat’s little theorem	C_{10}
9	Graphs: definitions, handshake lemma, paths/cycles, trees, Eulerian circuits, basic coloring	C_{11}
10	Graph enrichment: structural proofs and chromatic polynomial via deletion–contraction	—
Finals	Cumulative final exam (also serves as last-chance retake)	—

Advanced objectives A_1 – A_6 open as the prerequisite material is covered and remain open through finals.

Attendance

Students are expected to attend lecture and are responsible for all material covered in class. The first attempt at each Core PC takes place in class, so attendance on PC days is especially important.

Lateness

Homework is formative practice and there is no late penalty; however, late homework submitted after a given deadline will not retroactively unlock additional retake attempts.

Academic Policies

This course adheres strictly to the university’s policies on the following:

- Academic Dishonesty
- Course Add/Drop Policy
- Course Withdrawal Policy
- Incomplete Grade Policy
- Grade Appeals
- Code of Conduct

Disability Accommodations

It is the University’s policy to provide a learning environment that provides all students with the opportunity to realize their full potential. To this end, the goal of the Office of Disability Resources ("ODR") is to assist students with disabilities in becoming self-sufficient, independent, and successful members of the University’s community and to provide students with disabilities with equal opportunity of access to university courses, programs, facilities, services, and activities.

Students requesting accommodations due to a disability at Drexel University need to request a current Accommodations Verification Letter (AVL) in the ClockWork Database before accommodations can be made. These requests are received by Disability Resources (DR), who then issues the AVL to the appropriate contacts. For additional information, visit the website, or contact the ODR office by email disability@drexel.edu, or by phone 215-895-1401.

How accommodations work in this course. Because grading is mastery-based, every student already receives multiple attempts at each objective, so the structure is itself fairly accommodating. Authorized accommodations apply to how each proficiency check is administered: extended-time accommodations are applied to every PC and retake, with the standard PC length scaled by your authorized multiplier; reduced-distraction-environment accommodations are arranged either as a separately scheduled proctored sitting or through the ODR testing service. If an accommodation affects submission deadlines, the same extension applies to the homework deadlines that unlock additional retake attempts. Any details not covered here are arranged individually once I receive your AVL — please send it as early in the term as possible, so the arrangements are in place before the first PC.

Changes to Syllabus

Any changes to this syllabus will be communicated in class and by email.